

Ensemble Method in the Theory of Irreversibility

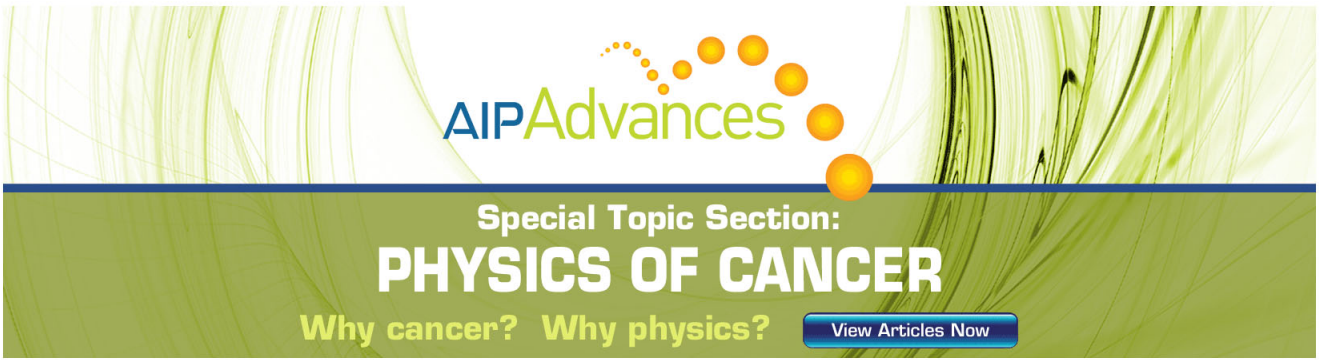
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formation can be obtained about their configuration in solution. Polynucleotides, polysaccharides, polypeptides, and proteins, for example, undergo changes in conformation upon variation of temperature, pH, solvent, etc. These changes, in general, do not involve any differences in primary chemical bonding, but they do give rise to changes in optical properties. These changes can be interpreted with the present equations. Differences between stereospecific and random synthetic polymers can also be treated. We should not ignore small molecules completely, however, one can make qualitative and perhaps quantitative predictions of the effect of substituents on the spectra of simple molecules also.

Other properties of a molecule which depend on the electron distribution can be considered from the present point of view. For all calculations, the information needed is the geometry of the molecule plus the dipole moment and transition moments of its constituent groups of electrons.

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Ensemble Method in the Theory of Irreversibility

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We describe a new formulation of methods introduced in the theory of irreversibility by Van Hove and Prigogine, with the purpose of making their ideas easier to understand and to apply. The main tool in this reformulation is the use of projection operators in the Hilbert space of Gibbsian ensemble densities. Projection operators are used to separate an ensemble density into a "relevant" part, needed for the calculation of mean values of specified observables, and the remaining "irrelevant" part. The relevant part is shown to satisfy a kinetic equation which is a generalization of Van Hove's "master equation to general order." Diagram summation methods are not used. The formalism is illustrated by a new derivation of the Prigogine-Brout master equation for a classical weakly interacting system.

INTRODUCTION

THIS article is the first of several to be devoted to the study, from a unified point of view, of various topics in the statistical mechanical theory of irreversibility.

Since the appearance in 1946 of Kirkwood's pioneering work in this field, activity has been intense and varied.

In the last few years significant advances have been made by Van Hove[1] and by Prigogine and his coworkers.[2] The work to be presented here originated in an effort to understand the methods introduced by these authors and to see their interrelations. In the course of this study it became evident that the Van Hove-Prigogine (VHP) methods can be made easier to understand and to use if a new point of view is taken. The initial formulation of this point of view is the main subject of this article. Future articles will deal with further developments.

We shall begin with a rough characterization of the VHP methods: (a) the initial state of a many-body system is specified; (b) the evolution of this state is followed by means of a many-body perturbation theory, involving expansion, rearrangement, approximation, and resummation of infinite series in powers of a perturbation; and (c) kinetic equations describing special features of this evolution are shown to be valid in an asymptotic (long-time) sense, for certain kinds of initial states.

An important feature of the VHP methods is that coarse-graining or time-smoothing is not needed. Random phase assumptions are not introduced except in specifying the initial state.

The examples worked out in detail by VHP are concerned with situations where a small parameter ξ (which may be a coupling strength, a density, etc.) characterizes the interaction responsible for irreversible behavior. They demonstrate the validity of kinetic equations in the limit t(time)$\rightarrow\infty$, $\xi\rightarrow 0$, and ξt is held constant. The formulation to be given in this article allows a simpler and more precise treatment of such weak interaction problems, but does not add anything substantially new to their results.

When the interaction parameter ξ is not small, the VHP methods become exceedingly complicated. The emphasis that has to be given to algebraic detail tends

[1] L. Van Hove, Physica **21**, 517 (1955); **25**, 268 (1959).

[2] I. Prigogine and R. Balescu, Physica **25**, 281, 302 (1959) and references given therein; R. Brout and I. Prigogine, *ibid.* **22**, 621 (1956).

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to obscure issues of a more physical kind. As a result it is hard to apply their methods and to assess the validity of their conclusions. The formulation to be presented here avoids some technical difficulties in the VHP methods. We hope that it will be useful in problems where ξ is not small, but at present no new results have been obtained in such problems.

Our reformulation of the VHP methods is characterized by extensive use of Gibbsian ensembles (in which we differ from Van Hove, whose concern with the evolution of initial pure quantum states is more in the Boltzmann tradition).

Projection operators in the space of all possible ensemble densities[3] are used to separate an ensemble density into a "relevant" part, required for the calculation of ensemble averages of specified quantities, and the remaining "irrelevant" part. This is a generalization of the common separation of a density matrix into diagonal and nondiagonal parts, as used for example in Van Hove's derivation of the Pauli equation.

The projected or relevant part of an ensemble density will be shown to satisfy a special kinetic equation. This equation is in some respects a generalization of Van Hove's "master equation to general order." If the initial ensemble density lies entirely within the subspace defined by the projection operator, our kinetic equation takes on a particularly useful form. It becomes a closed equation for the projected ensemble density; it describes the decay of this density; and it is not Markoffian.[4] This equation is of fundamental importance in the theory of irreversibility.

As an illustration of our formalism we shall present another derivation of a master equation, first derived by Prigogine and Brout,[2] for a weakly interacting system in classical mechanics.

DERIVATION OF A KINETIC EQUATION

Liouville's equation is the natural starting point of any theory of time-dependent processes in statistical mechanics. In this section it will be used in the derivation of a useful kinetic equation.

In order to avoid duplication, we shall introduce a notation which can be used simultaneously for both the classical and the quantum mechanical theory.

The ensemble density (phase space distribution function or density matrix) at time t is $f(t)$. The average of a dynamical variable α (function, matrix or operator) at time t is $\langle \alpha; f(t) \rangle$. The Hamiltonian function or operator is called H.

In classical statistical mechanics, Liouville's equation is conveniently written

$$i[\partial f(t)/\partial t] = i[H, f(t)] \equiv Lf(t), \tag{1}$$

where $[A, B]$ denotes the Poisson bracket of A and B. This equation defines the operator L, which is Hermitian by virtue of the imaginary unit i. Equation (1) has the formal solution

$$f(t) = \exp(-itL) f(0). \tag{2}$$

The operator $\exp(-itL)$, introduced by Von Neumann and used by Kirkwood, is defined by Eq. (2). It has the property of displacing a function in time along a trajectory in phase space.

In quantum statistical mechanics Liouville's equation takes the form

$$i[\partial f(t)/\partial t] = \hbar^{-1}[H, f(t)] \equiv Lf(t), \tag{3}$$

where $[A, B]$ is the commutator $AB-BA$. This equation defines a Hermitian operator L. Equation (3) has the formal solution

$$f(t) = \exp(-itL) f(0), \tag{4}$$

which is actually equivalent to

$$f(t) = \exp(-itH/\hbar) f(0) \exp(+itH/\hbar). \tag{5}$$

The solution in the form of Eq. (4) is preferred because of its analogy with the classical case. (Fano[5] has shown how to construct the quantum mechanical L explicitly).

In the rest of this section we shall not distinguish between the classical and quantum mechanical interpretations of this notation.

Now we introduce an operator P. We have in mind specifically a projection operator, but its projection property will not be used until later. P is used to divide an ensemble density $f(t)$ into a "relevant" part $f_1(t) = Pf(t)$ and an "irrelevant" part $f_2(t) = (1-P)f(t)$,

$$f(t) = f_1(t) + f_2(t). \tag{6}$$

We require that P be a linear operator and time independent so that we can commute P and $\partial/\partial t$. Liouville's equation can then be written as a pair of equations,

$$P[i(\partial f/\partial t)] = i(\partial f_1/\partial t) = PL(f_1+f_2), \tag{7a}$$

$$(1-P)[i(\partial f/\partial t)] = i(\partial f_2/\partial t) = (1-P)L(f_1+f_2). \tag{7b}$$

The second of these equations can be solved formally for $f_2(t)$ in terms of $f_2(0)$ and $f_1(t)$. In this solution we require the operator $\exp[-i(1-P)Lt]$ which is defined by the formal solution of

$$i(\partial G/\partial t) = (1-P)LG; \tag{8}$$

$$G(t) = \exp[-it(1-P)L] G(0). \tag{9}$$

The solution of Eq. (7b) is

$$f_2(t) = \exp[-it(1-P)L] f_2(0) - i\int_0^t ds \exp[-is(1-P)L](1-P)Lf_1(t-s). \tag{10}$$

[3] "Ensemble density" is used to denote either a quantum mechanical density matrix or a classical phase space distribution function.

[4] A kinetic equation is called Markoffian if it involves properties of an ensemble density only at a single time.

[5] U. Fano, Revs. Modern Phys. **29**, 74 (1957).

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When this is substituted in Eq. (7a) we obtain

$$i[\partial f_1(t)/\partial t]=PL\exp[-it(1-P)L]f_2(0)+PLf_1(t)$$
$$-i\int_0^t ds PL\exp[-is(1-P)L](1-P)Lf_1(t-s). \quad (11)$$

This is a generalization of Van Hove's "master equation to general order in the perturbation." It corresponds to his equation when (a) the Hamiltonian is separated into an unperturbed part and a perturbation, (b) we use the matrix representation in which the unperturbed Hamiltonian is diagonal, and (c) the projection operator selects the diagonal part of any matrix in this representation. Van Hove's interference terms are contained in $f_2(0)$.

Equation (11) is of particular value when $f_2(0)=0$, since it then becomes a closed equation for $f_1(t)$. The assumption $f_2(0)=0$ is equivalent to $Pf(0)=f(0)$. The projection operator P specifies a class of initial ensemble densities for which the assumption is valid.

The physical significance of the characterization $f_2(0)=0$ of a class of initial states will be discussed in a future article. We shall then describe a method of expanding an ensemble density in terms of a complete set of functions in Hilbert space. The basis functions are found by orthogonalizing a set of observables with respect to a microcanonical density. This method is an extension of one developed by Fano.[5] It may be used for the construction of initial ensemble densities in terms of prescribed information.

In Van Hove's derivation of the master equation, the assumption $f_2(0)=0$ is the same as discarding interference terms. It corresponds to an initial phase-randomization.

Equation (11) [henceforth with $f_2(0)=0$] has a number of interesting properties.

It is a closed equation for the projected ensemble density, giving correctly the time dependence of that part $f(t)$ which lies in the P-subspace, but giving no other information. If P projects onto the subspace of functions of dynamical invariants, then $\partial f_1(t)/\partial t=0$.

Equation (11) contains a memory, represented by the convolution in time. It describes a non-Markoffian process.

Another feature of Eq. (11) is its "irreversibility." Roughly speaking, the total ensemble density, initially in a certain subspace, "leaks" out of this subspace so that information is lost.

It is interesting to see the form taken by Eq. (11) under Fourier transformation. The transform of $f_1(t)$ is defined by

$$g(\omega)=\int_0^\infty dt\exp[i(\omega+i\epsilon)t]f_1(t), \quad (12)$$

where the frequency is given a positive imaginary part, $\epsilon>0$, to insure convergence. At the end of a calculation the limit $\epsilon\to 0^+$ should be taken. We consider the particular but typical case where the Hamiltonian separates into an unperturbed part H_0 and a perturbation H_1, and the operator L separates into L_0+L_1. Furthermore, we assume that the projection operator commutes with L_0, which is a common situation in practice. Then Eq. (11) can be written in terms of a "scattering operator" $T(\omega)$,

$$(\omega+i\epsilon)g(\omega)=if_1(0)+P(L_0+T(\omega))g(\omega), \quad (13)$$

where $T(\omega)$ satisfies an equation of the Lippmann-Schwinger[6] type,

$$T(\omega)=L_1+L_1(\omega+i\epsilon-L_0)^{-1}(1-P)T(\omega). \quad (14)$$

The kinetic theory of gases can be formulated using Eqs. (13) and (14), together with a binary collision expansion of $T(\omega)$. This will be reserved for a future article.

ILLUSTRATION

As an illustration of how the preceding formalism can be applied, we shall derive the Prigogine-Brout master equation in the limit of weak interaction.

We consider a classical system of N particles in a volume V. The Hamiltonian is

$$H=\sum_{j=1}^N (p_j^2/2m)+\tfrac{1}{2}\lambda\sum_{j\neq k=1}^N\sum U(r_{jk}), \quad (15)$$

where the jth particle has momentum p_j, position r_j, and mass m. The potential energy of interaction of the pair (jk) is $U(r_{jk})$ and depends only on their separation. The parameter λ determines the strength of the interaction.

The corresponding Liouville operator is

$$L=L_0+\lambda L_1,$$
$$L_0=-i\sum_{j=1}^N (p_j/m)(\partial/\partial r_j),$$
$$L_1=-(i/2)\sum_{j\neq k}\sum F_{jk}[(\partial/\partial p_j)-(\partial/\partial p_k)], \quad (16)$$

where

$$\lambda F_{jk}=-\lambda[\partial U(r_{jk})/\partial r_j] \quad (17)$$

is the force on j due to k.

The master equation is concerned with the momentum dependence of a phase space distribution function $f(r, p; t)$. We introduce the projection operator

$$P\equiv(V^N)^{-1}\int_V d^{3N}r, \quad (18)$$

which has the effect of removing all position dependence from f,

$$Pf(r, p; t)=f_1(p; t). \quad (19)$$

We shall consider only initial states which are spatially homogeneous, so that $Pf(0)=f(0)$ and $f_2(0)=$

[6] B. Lippman and J. Schwinger, Phys. Rev. **79**, 469 (1950).

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0. Then Eq. (11) becomes a closed equation for the spatially homogeneous part of the entire $f(t)$. The projected function $f_1(t)$ gives correct averages of all functions of momentum.

It is easy to show that the term $PLf_1(t)$ in Eq. (11) vanishes. The contribution from L_0 vanishes because f_1 is independent of position. The L_1 part is

$$PL_1f_1=-\tfrac{1}{2}i\sum_{j\neq k}\sum(V^N)^{-1}\int_V d^{3N}r F_{jk} \times[(\partial/\partial p_j)-(\partial/\partial p_k)]f_1(p;t). \quad (20)$$

This is zero because the integral of a force over all positions is zero. The same argument serves to eliminate a term $PLf_1(t-s)$ from Eq. (11). We are left with the exact kinetic equation

$$\partial f_1(t)/\partial t=-\int_0^t ds\,PL\exp[-is(1-P)L]Lf_1(t-s). \quad (21)$$

Next we use $Lf_1=\lambda L_1f_1$, since $L_0f_1=0$. We also note that (for arbitrary G)

$$PLG=\lambda PL_1G \quad (22)$$

because P removes the contribution from L_0. This gives

$$\partial f_1(t)/\partial t=\lambda^2\int_0^t ds K(s;\lambda)f_1(t-s), \quad (23)$$

where $K(s;\lambda)$ is an operator in momentum space,

$$K(s;\lambda)=-PL_1\exp[-is(1-P)L]L_1. \quad (24)$$

Equation (23), which is still exact, can be simplified further in the limit of weak interaction and long time. We introduce the new time scale,

$$\tau=\lambda^2t \quad (25)$$

and replace $f_1(t)$ by $\rho(\tau)$. Equation (23) becomes

$$\partial\rho(\tau)/\partial\tau=\int_0^{\tau/\lambda^2} ds K(s;\lambda)\rho(\tau-\lambda^2s). \quad (26)$$

We assume that the memory of $K(s;\lambda)$ extends only over a finite interval of time $0<s<T$ in the original time scale, or

$$K(s;\lambda)\cong 0 \quad \text{if} \quad s>T. \quad (27)$$

In actual cases the memory extends roughly over the time required for a particle of momentum p to traverse the distance over which the interaction does not vanish.

Now we can take the limit $\lambda\to 0$ at fixed $\tau>0$. In this limit we obtain

$$\partial\rho(\tau)/\partial\tau=\int_0^\infty ds K(s;0)\rho(\tau) \quad (28)$$

which is a Markoffian equation. The memory operator in the limit is

$$K(s;0)=-PL_1\exp(-isL_0)L_1 \quad (29)$$

because

$$(1-P)L=L_0+\lambda L_1-PL; \quad (30)$$

and PL is of order λ because of Eq. (22). Equations (28) and (29) are, in rather condensed notation, exactly the Prigogine-Brout master equation.

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